AISAP Frontend Challenge

Requirements - Core Stack

- Create a web application that simulates a lightweight echocardiography (Echo) study review dashboard.
- The application should allow users to browse cardiac echo studies, review their details, filter them, and update their status.
- A complete solution consists of:
  - Backend endpoints
  - A React-based frontend
  - Dockerized setup
  - You may implement the server side using a language/framework of your choice.

Functional Goals

- Attached static `studies.json` file that you'll use as data source for your application. No external APIs are required. Provide minimal backend with only one endpoint that get all studies.
- You required to implement the following in 3 separate pages/panels:

1. Fetch and Display Studies - Each study includes:
   - Patient name
   - Patient ID
   - Study date
   - Indication
   - LVEF (Left Ventricular Ejection Fraction)

2. Study Details View - Selecting a study should display detailed information in a separate panel or section. Details should include:
   - Patient information
   - Study date
   - Indication
   - LVEF value

3. Filtering & Search
   - Searching by patient name
   - Filtering by LVEF category:
     - Normal (≥55%)
     - Mildly reduced (40–54%)
     - Severely reduced (<40%)
   - Filtering must not reload the page.

- Pagination or Infinite Scroll - The dataset contains enough records to support pagination or infinite scrolling. Implement one of the following:

    - Client-side pagination
    - Infinite scroll
    - Load-more button

- Build a **responsive, user-friendly interface,** avoid unnecessary calls for backend.
- Show loading states while fetching data, error handling, and any relevant notifications.
- Your solution should be based on Docker containers. You may use:
    - Separate containers for frontend and backend
    - docker-compose for orchestration
- You can use AI tools as you want, you'll need to explain the frontend implementation in details.

Deliverables

- It is more important to take good design decisions than to submit a fully working solution.
- Please be ready to explain your decisions.
- Please provide full installation instructions for your solution.
- Submit your solution to a private Git repository and share it with hareld@aisap.ai, tal.kisra@aisap.ai, itay.tal@aisap.ai

Extra Credit

- Consider using either GitLab or GitHub.
    - Create a pipeline that builds your assets.

Good Luck

AISAP Team